

# **Exercise 1: Color Spaces and Image Bit Structure**

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Course: Image Processing

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# Contents

|          |                                         |          |
|----------|-----------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                     | <b>3</b> |
| <b>2</b> | <b>Part 1: HSV Color Space Analysis</b> | <b>3</b> |
| 2.1      | Requirements . . . . .                  | 3        |
| 2.2      | Expected Output Format . . . . .        | 3        |
| <b>3</b> | <b>Part 2: Bit-Plane Slicing</b>        | <b>4</b> |
| 3.1      | Requirements . . . . .                  | 4        |
| 3.2      | Expected Output Format . . . . .        | 4        |
| <b>4</b> | <b>Technical Notes &amp; Hints</b>      | <b>4</b> |
| <b>5</b> | <b>Input Description</b>                | <b>5</b> |
| <b>6</b> | <b>Submission Requirements</b>          | <b>5</b> |
| 6.1      | Naming Convention . . . . .             | 5        |

# 1 Introduction

In this exercise, we aim to become familiar with two key concepts in image processing: first, **separating color channels in HSV color space** – which is widely used for color-based segmentation – and second, **Bit-plane Slicing** – which helps us understand the role of each bit in the digital representation of images.

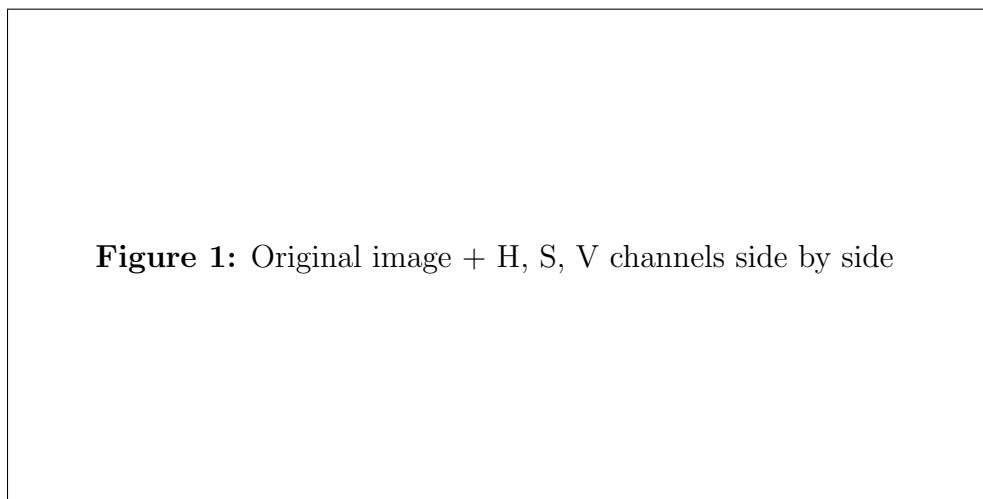
## 2 Part 1: HSV Color Space Analysis

The HSV color space, unlike RGB, separates **Hue**, **Saturation**, and **Value** (brightness) into independent channels. This makes operations like filtering a specific color much easier.

### 2.1 Requirements

1. Read the input image using OpenCV.
2. Convert the image to HSV color space.
3. Separate the three channels (H, S, V) from each other.
4. Display all three channels along with the original image in **one figure** (using `plt.subplot`). Use `cmap='gray'` for displaying the channels.
5. Looking at the outputs, **explain why** saturated (thick/colorful) parts of the image appear **brighter** in the S channel.

### 2.2 Expected Output Format



**Figure 1:** Original image + H, S, V channels side by side

Figure 1: HSV color space visualization

### 3 Part 2: Bit-Plane Slicing

Grayscale images (8-bit) consist of 8 binary layers. **Bit 7 (MSB)** contains the overall structure of the image, while **Bit 0 (LSB)** contains very fine details and noise.

#### 3.1 Requirements

1. Load the input image in **grayscale**.
2. Using **bitwise operations in NumPy** (e.g., the **&** operator), decompose the image into **8 bit-planes**.
3. Display all 8 bit-planes in an organized layout (e.g., 2×4 grid).
4. In your report, mention **which bits can be removed** while the image remains **recognizable to the human eye**.

#### 3.2 Expected Output Format

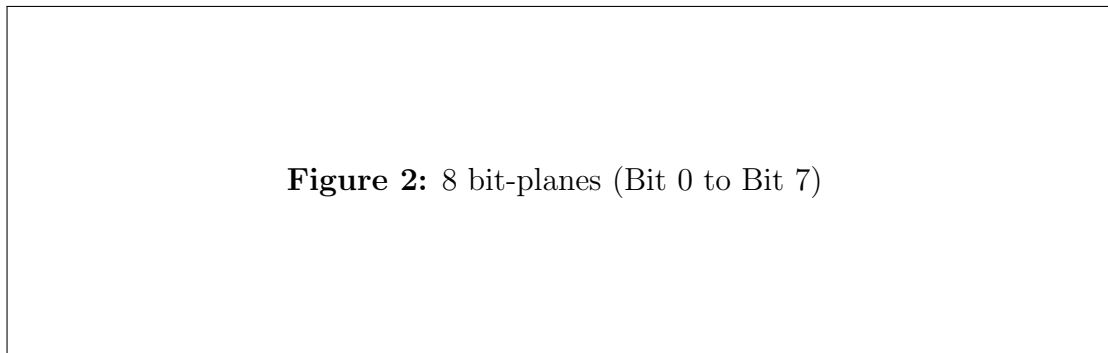


Figure 2: Bit-plane decomposition results

### 4 Technical Notes & Hints

| Item              | Description                                                                     |
|-------------------|---------------------------------------------------------------------------------|
| Input file        | Name the input image exactly as given in the attached file                      |
| Allowed libraries | <code>cv2</code> , <code>numpy</code> , <code>matplotlib.pyplot</code>          |
| Display           | Use appropriate <code>figsize</code> to ensure outputs are clear in your report |
| Bit order         | Bit 7 = Most Significant Bit (MSB), Bit 0 = Least Significant Bit (LSB)         |

Table 1: Technical specifications

## 5 Input Description

*[Description of the input image – fill in as provided by your instructor]*

- **Image format:** \_\_\_\_\_
- **Image dimensions:** \_\_\_\_\_
- **Color space:** RGB (for Part 1), will be converted (for Part 2)

## 6 Submission Requirements

Please upload a **compressed file (.zip)** containing:

1. **Implementation code** (as .py or .ipynb file)
2. **A PDF file** including:
  - Output images (figures you extracted)
  - Answers to the short questions in each section

### 6.1 Naming Convention

IP\_StudentName\_StudentId.zip

**Good luck!**